

Question Number	Answer	Additional Guidance	Mark
4a	A description that includes two of the following points: • find a suitable {mass / length} of pondweed (1) • find a suitable temperature/pH (for photosynthesis) (1) • find suitable method to measure the {oxygen / gas} production (1) • find suitable depth(s) of water or light intensity (1) • find a suitable method to change depth of water or light intensity (1)	A method to provide quantitative results Allow time to measure gas production Allow to change wavelength of light	2

	Answer	Additional Guidance	Mark
4b	An answer that includes nine of the following points: • clear statement of the dependent variable e.g. the volume of {gas / O₂} released per unit time (1) • description of method of measuring volume of gas (1) • method of standardising plants (1) • method of producing two (or more) different depths of water or different light intensities (1) • use of (sodium) hydrogencarbonate (1) • allow plant time to acclimatise (1) • {standardised / stated} time for gas collection (1) • one variable that need to be controlled and its method of control (1) • method of calculating rate (1) • {repeats / repeat the whole experiment} and calculate {means / SD's} (1)	Ignore amount / bubble /rate Allow distance moved per unit time Allow photosynthometer / labelled diagram Allow mass /length Ignore age Allow equilibrate Allow 10 mins to 24hrs (if given) e.g. temp / pH / wavelength / light intensity (if depth is changed) / depth (if light intensity is changed) e.g. distance ÷ time vol ÷ time number of bubbles ÷ time	9

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4c	An answer that includes the following points: - table for raw data with headings and units, with means calculated from repeats (1) AND - bar graph format with labelled axes (1) - use of a t test for difference (with a bar graph showing only two different light intensities) (1) OR {line / scatter} graph format with labelled axes (1) - use of a correlation test (with a line graph showing more than two different light intensities) (1)	Reject rate in heading Allow description of mean calculated in text or mean on one graph label Graph must be clearly sketched or stated Allow named test Graph must be clearly sketched or stated Allow named test	3

